

of coherent detector + more complexity.

Module 02:- DSBSC Demodulation:-

There are two different techniques

- i) Coherent Detection
- ii) Costas receiver

1. Coherent Detection of DSBSC wave

The modulating sig $[m(t)]$ is obtained from DSBSC modulated signal $S(t)$ by first multiplying $S(t)$ with locally generated carrier signal $\cos(2\pi f_c t + \phi)$ and then low pass filtering the product as shown in below figure.

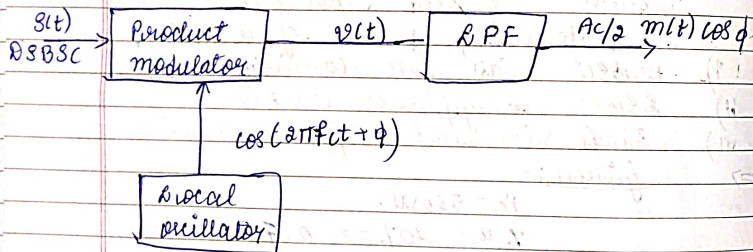


Fig. 1: Coherent detector

The output of a product modulator is given by

$$v(t) = S(t) \cdot \cos(2\pi f_c t + \phi) \quad \text{--- (1)}$$

W.S.T

$$S(t) = A_c \cos 2\pi f_c t \cdot m(t) \quad \text{--- (2)}$$

Substitute (2) in (1).

We get,

$$v(t) = A_c \cos 2\pi f_c t \cdot m(t) \cdot \cos(2\pi f_c t + \phi) \quad \text{--- (3)}$$

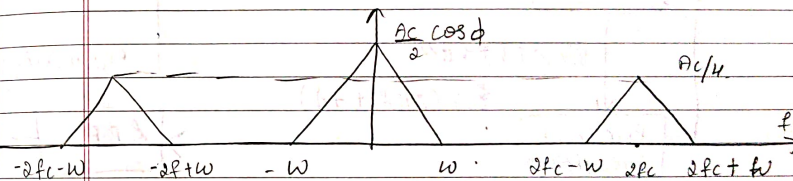
The above eqⁿ can be simplified using $\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$

$$v(t) = \frac{A_c \cos(\phi)}{2} m(t) + \frac{A_c \cos(4\pi f_c t + \phi)}{2} m(t) \quad \text{--- (4)}$$

Taking Fourier Transform of a resulting eqⁿ we get

$$V(f) = \frac{A_c}{2} [M(f-\phi) + M(f+\phi)] + \frac{A_c}{4}$$

$$V(f) = \frac{A_c}{2} M(f) \cos \phi + \frac{A_c}{4} [M(f-2f_c) + M(f+2f_c)]$$



The message sig $m(t)$ is obtained by passing $v(t)$ through Low Pass Filter having a bandwidth greater than W Hz but less than $2f_c - W$. The o/p of Low Pass filter is given by.

$$v(t) = \frac{A_c}{2} m(t) \cos \phi$$

where ϕ = phase error.

* When ϕ = constant, The filter o/p is proportional to $m(t)$.

* When $\phi = 0$, the amplitude of filter o/p is maximum i.e., $A_c/2$.

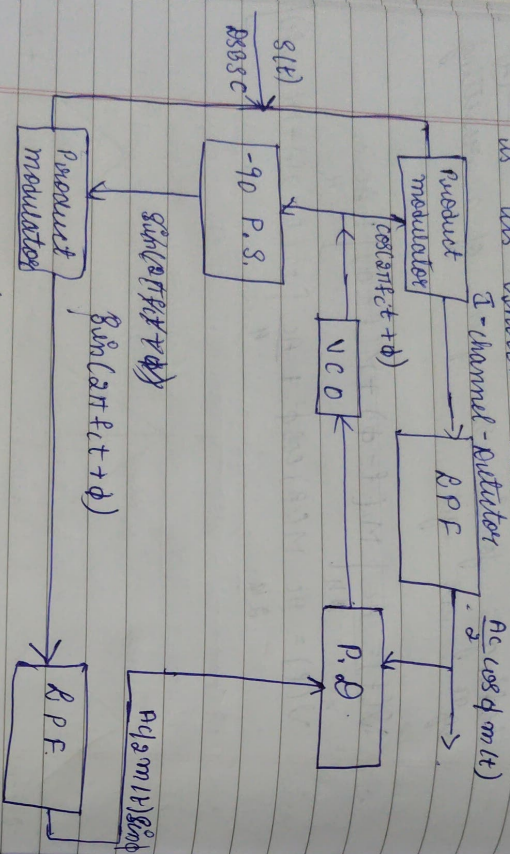
* When $\phi = \pm \pi/2$, the amplitude of filter o/p is zero.

10 marks

O/P is minimum i.e., 0.

Costas Receiver / Loop / Practical Synchronous Receiver system.

The block diagram of Costas receiver is as shown below.



8-channel detector

It consists of two coherent detectors operating with common I/P i.e., incoming BSC sig but with individual local oscillator signals which are in phase quadrature w.r.t each other. But with ~~in the~~ i.e.,

frequency of local oscillators is adjusted to same as carrier sig f_c

The detector of upper path is suspended as a inphase detector (or) I-channel detector

Similarly the detector in the lower path

is suspended as the quadrature detector / 8-channel detector. The two detectors are connected together to form -ve feedback system. This system is designed in such a way that it maintains the local oscillator synchronous with carrier sig.

Equation:-

To understand the ckt opⁿ of this receiver, let us consider that the local oscillator sig is ~~same~~ ϕ in the same phase as the carrier sig. At receiver side under this condⁿ the I-channel O/P is given by $\frac{A_c}{2} \cos(\phi) \cos(\phi) = \frac{A_c}{2} \cos(2\phi)$

If the P.D. O/P is directly proportional $\sin(\phi)$ but the lower channel O/P i.e., 8-channel O/P $\frac{A_c}{2} \sin(\phi) \sin(\phi) = 0$

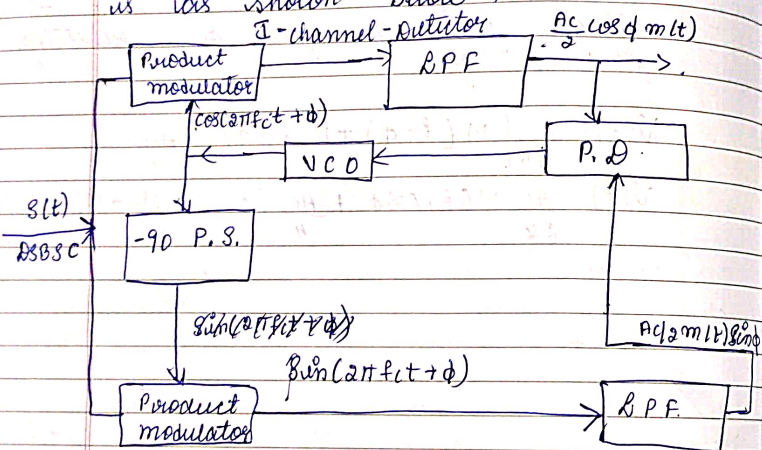
We consider when the local oscillator phase shift from its proper value by a very small angle ϕ explains. Under this condition the I-channel O/P remains same i.e., $\frac{A_c}{2} \cos(\phi) \cos(\phi)$. But the 8-channel produces some O/P i.e., proportional to $\frac{A_c}{2} \sin(\phi) \sin(\phi)$

The O/P of 8-channel & I-channel are combined in phase discriminator which consists of multiplier followed by loopous filter. This device converts the phase difference into a DC control-voltage.

o/p is minimum i.e., 0.

Costas Receiver / Loop / Practical Synchronous Receiver System:

The block diagram of Costas receiver is as shown below.



Q-channel detector

- It consists of two coherent detectors operate with common I/P i.e., incoming DSBSC sig. but with individual local oscillator signals which are in phase quadrature w.r.t each other. i.e., 90° phase shift.
- Frequency of local oscillator is adjusted to same as carrier sig f_c .
- The detector of upper path is referred as a inphase detector (or) I-channel detector.
- Similarly the detector in the lower path

is referred as the quadrature detector / Q-channel detector.

- The two detectors are connected together to form a feedback system.
- This system is designed in such a way that it maintains the local oscillator synchronous with carrier sig.

Operation:-

- To understand the ckt operⁿ of this receiver, let us consider that the local oscillator sig is ϕ in the same phase as the carrier sig $A \cos 2\pi f_c t$ used at the transmitter side.
- Under this condⁿ the I-channel o/p is given by $\frac{A_c}{2} m(t) \cos \phi$ (1) = $\frac{A_c}{2} m(t)$.

For the p.e., o/p is directly proportional $m(t)$, but the lower channel o/p i.e., Q-channel o/p $\frac{A_c}{2} \sin \phi m(t) = 0$.

- We consider when our local oscillator phase shift from its proper value by a very small angle ϕ radians. Under this condition the I-channel o/p remains same i.e., $\frac{A_c}{2} m(t) \cos \phi$. But the Q-channel produces some o/p i.e., proportional to $\frac{A_c}{2} m(t) \sin \phi$.

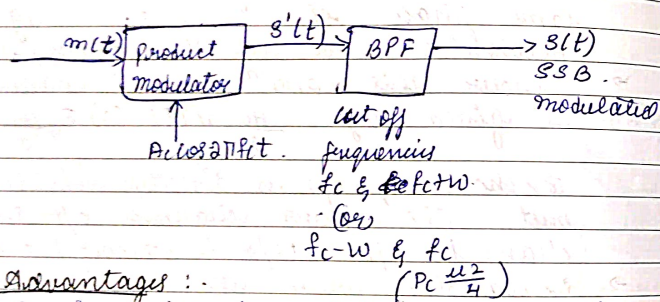
- The o/p of Q-channel & I-channel are combined in phase discriminator which consists of multiplier followed by low pass filter.

- This device converts the phase difference into a DC control voltage.

→ This control voltage automatically corrects the local oscillator phase error in voltage control oscillator.

* SSB Modulation:-

The generation of SSB modulated sig is very straight forward. We first generate DSBSC modulated sig then apply resulting DSBSC sig to a bandpass filter as illustrated below.



* Advantages:-

- Requires less transmission power and transmission bandwidth is equal to w .
- The major application of SSB modulation is transmission of analog voice sig. This analog voice sig has a very little energy at low frequency range ($< 300\text{ Hz}$) as illustrated below.

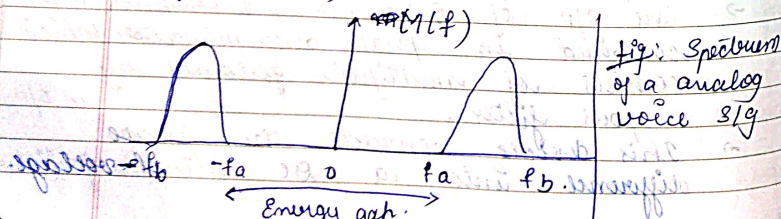
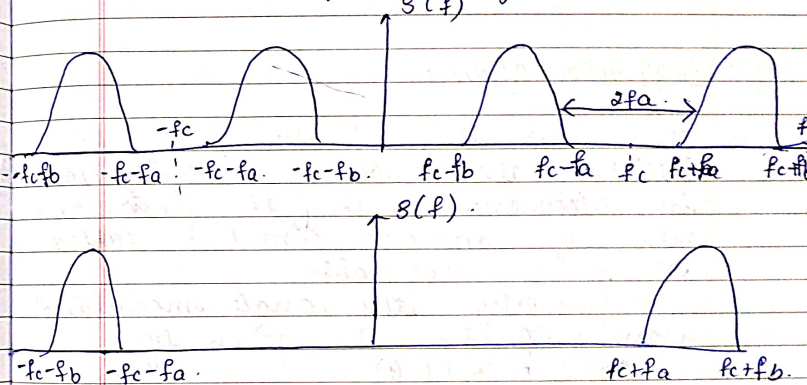


Fig. Spectrum of a analog voice sig

- It has energy gap near the origin.
- This analog voice signal is multiplied with carrier signal $A_c \cos 2\pi f_c t$ to generate DSBSC signal. $s'(t) = A_c \cos 2\pi f_c t * m(t)$

By taking Fourier transform of it we get $s'(f) = \frac{A_c}{2} [M(f-f_c) + M(f+f_c)]$

Hence the spectrum of DSBSC sig is as below for the given analog.



- + Passing the resulting DSBSC sig through a bandpass filter with cut off frequencies $f_c + f_a$ & $f_c + f_b$ we can retain USB at o/p of a filter. The spectrum of a resulting SSB sig is shown as above.

* Angle Modulation:-

* Comparison b/w SSB AM, DSBSC, SSB

	SSB AM	DSBSC	SSB
No. of side band components	USB LSB Carrier	Two USB & LSB	One SSB (or) USB
Required Bandwidth	2W Hz	2W Hz	W Hz
Total Transmitted Power P_t	$P_t = P_c(1 + \mu^2/2)$	$P_t = P_c \mu^2/2$	$P_t = \frac{P_c \mu^2}{4}$

1. Angle modulation :-

→ Angle modulation is the process of altering the angle of carrier s/g in accordance with the instantaneous value of message signal is called as angle modulation.

Generally the angle modulated wave equation is given by

$$s(t) = A_c \cos[\phi(t)]$$

Where A_c = peak amplitude of carrier s/g
 $\phi(t)$ = angle of modulated s/g

→ Angle modulation technique is further divided into 2 types -
i) Frequency modulation
ii) Phase modulation

i) Frequency Modulation :- Process of altering the instantaneous value of frequency of carrier s/g w.r.t message s/g keeping other two parameters constant such as

$$\text{mod } \omega \rightarrow K_f m(t)$$

$$\text{mod } \omega \rightarrow K_f$$

amplitude and phase of a carrier signal is constant

→ The general eqⁿ of a FM signal is given by
 $s(t)_{FM} = A_c \cos[2\pi f_c t + 2\pi K_f \int m(t) dt]$

ii) Phase Modulation :- Process of altering phase/angle of carrier s/g w.r.t message s/g keeping the other two parameters such as amplitude and frequency of a carrier signal as constant.

→ In general the phase modulated wave eqⁿ is given by
 $s(t)_{PM} = A_c \cos[2\pi f_c t + K_p m(t)]$

1. Basic Definitions

i) Instantaneous Frequency :-

A complete oscillation occurs whenever $\phi(t)$ changes by 2π radians. If $\phi(t)$ increases w.r.t time t , the average frequency over the time interval from t to $t + \Delta t$ is given by

$$f_{\text{avg}}(t) = \frac{\phi(t + \Delta t) - \phi(t)}{2\pi \Delta t}$$

Inst frequency is equal to avg frequency
i.e., $f(t) = \lim_{\Delta t \rightarrow 0} f_{\text{avg}}(t)$

$$= \lim_{\Delta t \rightarrow 0} \frac{\phi(t + \Delta t) - \phi(t)}{2\pi \Delta t}$$

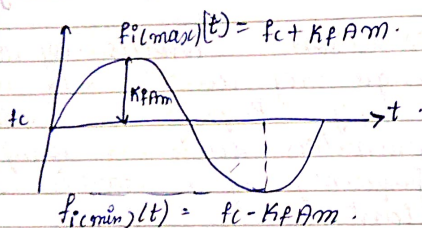
$$f(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

①

Band in the definition of FM
the instantaneous frequency
is defined as follows.

$$f_i(t) = f_c + K_f m(t)$$

$$f_i(t) = f_c + K_f m(t)$$



2) Frequency Deviation:-

It is the difference b/w
max instantaneous frequency and carrier
frequency.

Frequency Deviation = max inst.

$f_{i(max)}(t)$ - carrier frequency

$$\Delta f = f_{i(max)}(t) - f_c$$

$$\Delta f = K_f A_m$$

where K_f = sensitivity of FM frequency
- b/w for FM

$$K_f = \frac{\Delta f}{A_m}$$

3) Carrier Swing:-

It is the difference b/w
the $f_{i(max)}(t)$ and $f_{i(min)}(t)$

Carrier swing = $f_{i(max)}(t) - f_{i(min)}(t)$

$$f_{i(max)}(t) = f_c + A_m K_f$$

$$f_{i(min)}(t) = f_c - A_m K_f$$

$$\text{Carrier swing} = 2 A_m K_f$$

$$= 2 \Delta f$$

$$\therefore \Delta f = A_m K_f$$

4) Modulation Index:- (β)

It is the ratio of
frequency deviation to the frequency
of the message signal.

$$\beta = \frac{\Delta f}{f_m}$$

Properties of angle modulated s/g:-

- ① Constancy of transmitted power.
- ② Non-linearity of modulation process.
- ③ Irregularity of zero crossing.
- ④ Visualization difficulty of message wave.
- ⑤ Trade off of increasing the transmission BW for improved noise performance.

① Constancy of Transmitted power:-

To prove this property, let us consider
angle modulated wave eqⁿ Equations.

$$s(t)_{FM} = A_c \cos[2\pi f_c t + 2\pi K_f \int m(t) dt]$$

$$s(t)_{PM} = A_c \cos[2\pi f_c t + K_p m(t)]$$

from these two Equations, we can
notice that the amplitude of
angle modulated wave eqⁿ is
maintained constant amplitude
and it is equal to A_c for all time t .
 $\therefore$ The total transmitted power is
given by $P_t = P_{avg} = \frac{A_c^2}{2}$ with $R_B = 1/2$

Since A_c is constant, the transmitted
is also constant.

② Non-linearity of Modulating Process:

In order to prove this property, let us consider the message s/g $m(t)$ is made up of two different components such as $m_1(t)$ and $m_2(t)$ i.e., $m(t) = m_1(t) + m_2(t)$. Let $S_1(t)$, $S_2(t)$ denote phase modulated s/g produced by $m_1(t)$, $m_2(t)$ respectively and thus modulated s/gs are given by

$$S_1(t) = A_c \cos[2\pi f_c t + k_p m_1(t)] \quad \text{--- ①}$$

$$S_2(t) = A_c \cos[2\pi f_c t + k_p m_2(t)]$$

$$\text{Compute } S_1(t) + S_2(t)$$

$$S_1(t) + S_2(t) = A_c [\cos(2\pi f_c t + k_p m_1(t)) + \cos(2\pi f_c t + k_p m_2(t))] \quad \text{--- ②}$$

$$\therefore S_1(t) \neq S_2(t) + S_3(t)$$

From ① & ② it can be noticed that $S_1(t) \neq S_2(t) + S_3(t)$

The above eqⁿ indicates that the angle modulation process does not obey the principles of superposition. Hence it is a non-linear process.

③ Irregularity of Zero Crossing:-

To prove this property, let us consider angle modulated wave eqⁿ

$$S(t)_{FM} = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t) dt]$$

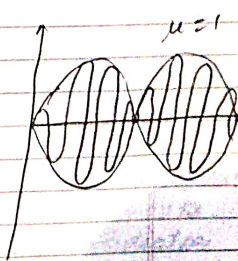
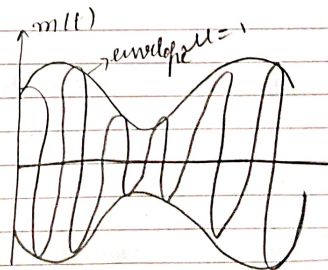
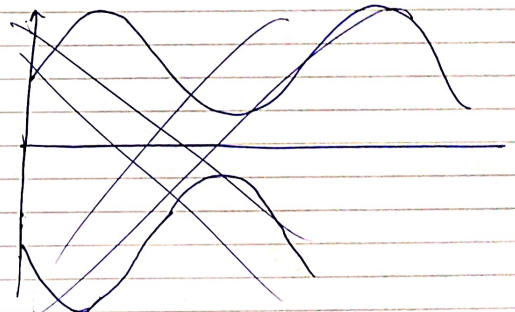
$$S(t)_{PM} = A_c \cos[2\pi f_c t + k_p m(t)]$$

From these two eqⁿ we can notice

that instantaneous angle $\phi(t)$ depends on the message signal $m(t)$. Hence FM wave of PM wave must be regular with $m(t)$ w.r.t their timeline. But in general the zero crossing of FM (or) PM wave don't perfectly regular with their timeline, hence they exhibit irregularity of zero crossing.

④ Visualization: Difficulty of message s/g

In AM we see that the message s/g $m(t)$ as an envelope of modulated s/g provided that modulation index $\mu \leq 1$ as illustrated below



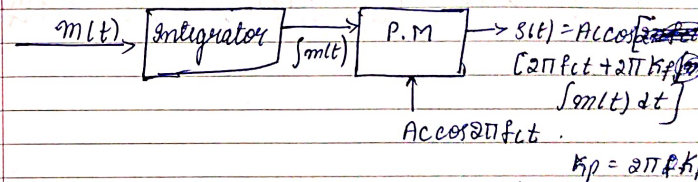
→ Envelope resembles the message signal
But PM or FM it is not possible. So.
In general the difficulty in
visualization of a message signal
in angle modulated waveform is
a attribute of non-linear
characteristics of a angle modulation
process.

③ Trade off Increasing Transmission Bandwidth
for improved noise performance :-

amp
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Generation of FM wave using phase Modulator

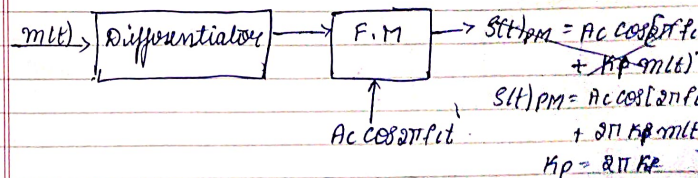
The scheme used to generate frequency modulated waveform using phase modulator is shown below.



This scheme consists of integrator followed by phase modulator. FM wave is generated by first integrating the message signal $m(t)$ and the resulting signal is applied to the phase modulator along with carrier signal $A \cos 2\pi f_c t$.

Generation of PM wave using Frequency Modulator

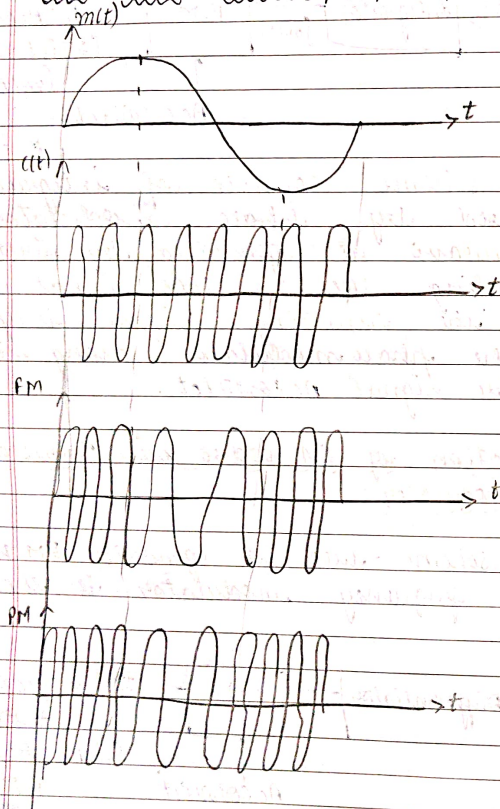
The scheme used to generate PM wave using frequency modulator is shown below.



This scheme consisting of differentiator

followed by frequency modulator
P-M phase modulated waveform is generated
by first differentiating the $m(t)$.
The resulting $m'(t)$ is applied
to the FM along with carrier
sig $A_c \cos 2\pi f_c t$.

PM and FM - input/output waveforms:-



Frequency modulation for single tone:-

For single tone, let us consider $m(t)$
message signal $m(t) = A_m \cos 2\pi f_m t$
where A_m = peak amplitude of $m(t)$
 f_m = frequency of $m(t)$.

The carrier signal $c(t)$ is given by
 $c(t) = A_c \cos 2\pi f_c t$

where $c(t)$ = inst value of carrier sig
 A_c = max amplitude of carrier
sig

In general the FM wave equation is
given by $s(t) = A_c \cos [2\pi f_c t + 2\pi K_f \int m(t) dt]$ — (1)

For single tone modulation

$$m(t) = A_m \cos 2\pi f_m t$$

$$\int m(t) dt = \int A_m \cos 2\pi f_m t dt$$

$$= A_m \frac{\sin 2\pi f_m t}{2\pi f_m} \quad \text{--- (2)}$$

Substitute (2) in (1)

$$s(t) = A_c \cos [2\pi f_c t + 2\pi K_f \frac{A_m \sin 2\pi f_m t}{2\pi f_m}]$$

where $K_f A_m = \Delta f$ = frequency deviation

$$s(t) = A_c \cos [2\pi f_c t + \frac{\Delta f}{f_m} \sin 2\pi f_m t]$$

where $\Delta f / f_m = \beta$ = modulation index

Hence FM wave for Eqⁿ for single
tone modulation is given by

$$s(t) = A_c \cos [2\pi f_c t + \beta \sin (2\pi f_m t)]$$

* Types of FM :-

Depending upon the value of β the ~~FM~~ ^{FM} wave is classified into 2 types -

- i) Narrow band FM for $\beta < 1$
- ii) Wide band FM for $\beta > 1$

11.3) Narrow Band FM

To derive the eqⁿ for narrow band FM let us consider the eqⁿ of FM for single tone modulation i.e. ~~FM~~

$$s(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]$$

The above eqⁿ is simplified using the trigonometric fun^c

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$s(t) = A_c \cos[2\pi f_c t] \cos[\beta \sin(2\pi f_m t)] - \sin[2\pi f_c t] \sin[\beta \sin(2\pi f_m t)]$$

w.k.T

$$\cos \theta = 1 \Rightarrow \theta = 0 \text{ very low.}$$

$$\sin \theta \approx \theta \Rightarrow \theta \text{ is very low.}$$

Since for narrow band FM $\beta < 1$, $\cos[\beta \sin(2\pi f_m t)] \approx 1$

$$\sin[\beta \sin(2\pi f_m t)] \approx \beta \sin(2\pi f_m t)$$

With this approximation $s(t)$ eqⁿ can be rewritten as follows.

$$\therefore s(t) \approx A_c [\cos 2\pi f_c t \cdot 1 - \sin(2\pi f_c t) \beta \sin(2\pi f_m t)]$$

$$\approx A_c \cos 2\pi f_c t - \beta A_c \sin(2\pi f_c t) \sin(2\pi f_m t)$$

The second term can be simplified using the trigonometric function

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$s(t)$ can be rewritten as

$$s(t) \approx A_c \cos 2\pi f_c t - \frac{\beta A_c}{2} \left\{ \cos[2\pi(f_c - f_m)t] - \cos[2\pi(f_c + f_m)t] \right\}$$

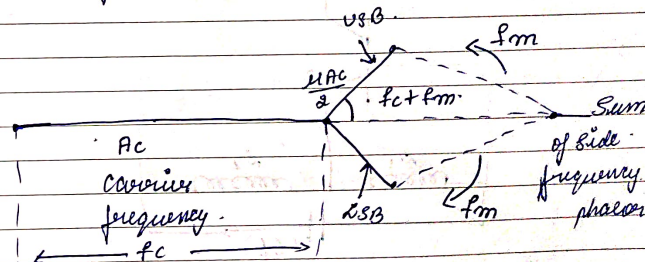
$$s(t) \approx A_c \cos 2\pi f_c t + \frac{\beta A_c}{2} \cos 2\pi(f_c + f_m)t - \frac{\beta A_c}{2} \cos 2\pi(f_c - f_m)t$$

- The resulting narrow band FM wave eqⁿ consists of following 3 components
- i) carrier component with the amplitude A_c and frequency f_c
 - ii) USB component with the amplitude $\beta A_c/2$ and frequency $(f_c + f_m)$
 - iii) LSB component with the amplitude $-\beta A_c/2$ and frequency $(f_c - f_m)$

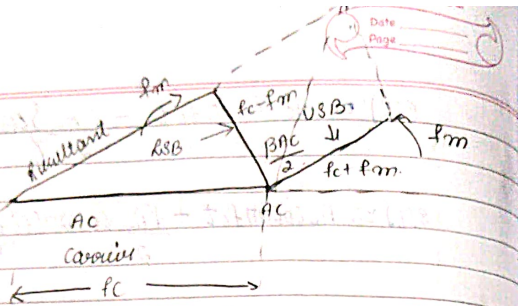
Let us consider AM wave equation for single tone modulation.

$$s(t) = A_c \cos 2\pi f_c t + \frac{A_c \mu}{2} [\cos 2\pi(f_c + f_m)t] + \frac{A_c \mu}{2} [\cos 2\pi(f_c - f_m)t]$$

The phasor diagram of AM s/g is illustrated as follows



The phasor diagram of FM s/g is illustrated as follows



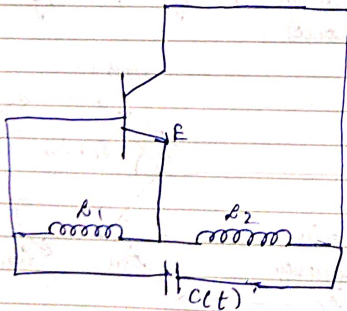
Generation of FM signal:-

There are two different approaches.

- Direct method
- Indirect method.

i) Direct Method of generating FM signal:-

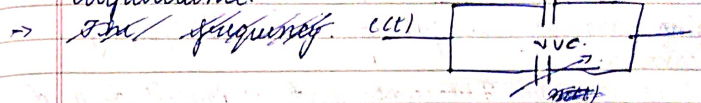
The direct method of generation of FM s/g uses sinusoidal high frequency sinusoidal oscillator ckt one shown below.



The above figure shows Hartley's oscillator in which the capacitive element of the tank ckt i.e., $C(t) = C_0 + \Delta C \cos(2\pi f_m t)$ where C_0 is total capacitance in absence

of $m(t)$

ΔC is the max change in the capacitance.



The frequency of Hartley is a oscillator is given by $f_i(t) = \frac{1}{2\pi \sqrt{(L_1 + L_2) C(t)}}$

In the absence of message s/g $C(t) = C_0$. Hence inst frequency of oscillator $f_i(t) = \frac{1}{2\pi \sqrt{(L_1 + L_2) C_0}}$

$$f_i(t) = \frac{1}{2\pi \sqrt{(L_1 + L_2) (C_0 + \Delta C \cos 2\pi f_m t)}}$$

$$= \frac{1}{2\pi \sqrt{(L_1 + L_2) C_0} \sqrt{1 + \frac{\Delta C \cos 2\pi f_m t}{C_0}}}$$

$$= \frac{f_0}{\left(1 + \frac{\Delta C \cos 2\pi f_m t}{C_0}\right)^{1/2}}$$

$$f_i(t) = f_0 \left(1 + \frac{\Delta C \cos 2\pi f_m t}{C_0}\right)^{-1/2}$$

$$\text{where } \left(1 + \frac{\Delta C \cos 2\pi f_m t}{C_0}\right)^{-1/2} \approx 1 - \frac{\Delta C \cos 2\pi f_m t}{2C_0}$$

$$f_i(t) = f_0 \left[1 - \frac{\Delta C \cos 2\pi f_m t}{2C_0}\right]$$

$$\frac{-\Delta C}{2C_0} = \frac{\Delta f}{f_0}$$

$$\Delta f(t) = f_0 \left[1 + \frac{\Delta f}{f_0} \cos 2\pi f_m t\right]$$

$$f_i(t) = f_0 + \Delta f \cos 2\pi f_m t$$

Comparison b/w AM and FM:

Parameter	AM	FM
changing parameter of a carrier	Amplitude	Frequency
Constant parameter of a carrier	Frequency and phase	Amplitude and phase
Modulation index	$\mu = \frac{A_m}{A_c} \leq 1$	$\beta = \frac{\Delta f}{f_m}$
		$\beta < 1$ [NBFM] $\beta > 1$ [WBFM]
Transmitted Power	$P_t = P_c \left[1 + \frac{\mu^2}{2} \right]$	$P_t = \frac{A_c^2}{2 R_L}$
Max Power Efficiency	33.33%	100%
Bandwidth	2W	2W if $\beta < 1$ $2\Delta f \left(1 + \frac{1}{\beta} \right)$ if $\beta > 1$ $\approx 2\Delta f + 2f_m$
Range of communication	Covers longer range	Short range
Complexity	Less complex	More complex
Immunity to noise	Less immunity + Affected by noise	More immunity + Less affected by noise
Applications	+ used for long distance comm	+ used for short distance comm

The equation for FM is given by
 $s(t) =$